GAVIN COUCH

Computer Science / Software Engineering

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Portfolio: gavincouch.github.io

PROJECTS

ApolLens

github.com/AdityaB2007/ApolLens

2nd place, Advanced category | Hackathon team project

- Developed the Flutter application for an offline repair assistant that identifies laptop components through the phone camera and provides step-by-step repair guidance.
- Trained and integrated a YOLOX-Nano component detector using ONNX Runtime; connected local language models to the repair flow.

Ordinary Days

github.com/GavinCouch/OrdinaryDays

Ren'Py 8.5.3 | Visual novel in development

- Created a school-life visual novel combining romance, comedy, and psychological horror, with dialogue choices, character reactions, scenes, sprites, and audio.
- Released a five-day Chapter 1 playtest for Windows x64 with saves, history, text and audio controls, Reduced Motion, and GitHub feedback workflows.

Health Easy

github.com/GavinCouch/Health-Easy

Swift 6 / SwiftUI | iPhone application

- Built food logging, saved meals, weight entries, and progress charts; integrated Apple and Google sign-in, Firestore persistence, and Firebase Storage for photos.
- Connected HealthKit active energy data and a recipe API; added photo comparisons with export and sharing.

TECHNICAL SKILLS

Languages and UI: Swift, SwiftUI, Dart, Flutter, Ren'Py
Data and models: Firebase, Firestore, HealthKit, ONNX Runtime, local language models
Tools and APIs: Git, GitHub, REST APIs, JSON, Swift Charts

EDUCATION

Louisiana State University (LSU)

Third-year student

Computer Science - Software Engineering

EXPERIENCE

Software Engineering Intern - Geek Squad — Upcoming winter internship

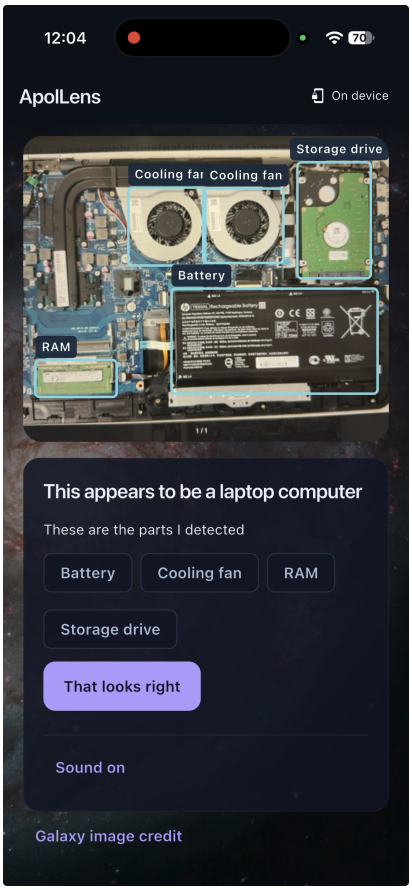
Advanced Repair Agent - Geek Squad — July 2025 - Present

PROJECT DETAILS

ApolLens

2nd place, Advanced category

Built around the challenge of maintaining equipment beyond Earth. The offline prototype identifies laptop components and provides visual and spoken repair guidance.



Laptop component detection

From photo to repair

Confirm detected parts, describe the problem, then follow repair steps with spoken guidance. The detector recognizes batteries, cooling fans, RAM, and storage drives.

My work

Flutter app, local language-model integration, computer-vision training and mobile integration, UI development, and device testing. Built as part of a hackathon team.

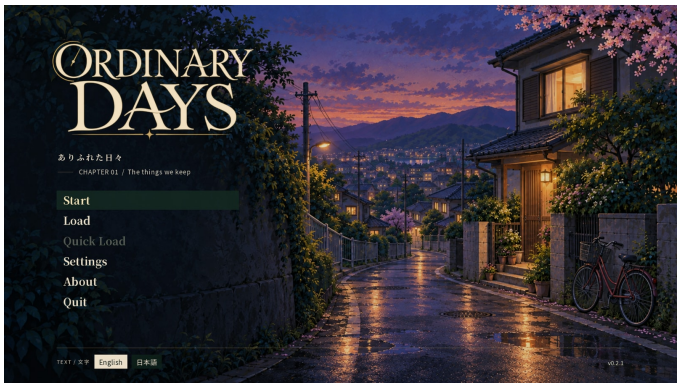
Implementation

Flutter / Dart; YOLOX-Nano and ONNX Runtime; Apple Foundation Models with Qwen3-1.7B via LiteRT-LM fallback; text-to-speech. Models and voices must be installed before offline use.

View team repository

Ordinary Days

A school-life visual novel where dialogue choices shape character reactions. Chapter 1 is available as a Windows playtest.

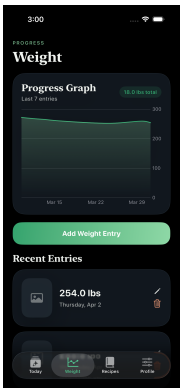


View project and playtest

Health Easy

An iPhone app for logging meals and weight, viewing progress, and comparing photos.





View app source